

Generation Interconnection Request System Impact Study Report

For Queue Project Q17-0725

Stuttgart Ricuskey 230 kV
January 29, 2018

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q17-0725. The Interconnection Customer proposes a 107.2 MW solar photovoltaic facility interconnecting at the Stuttgart Ricuskey 230 kV substation in the Illinois area. The request was received on January 14, 2017. The study evaluated system intact and single contingency conditions using the regional planning case, including 335 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 3 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$563,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

1. Facility: Canal Road – Weedpatch 230.0 kV
Contingency: Trego – Beaver 500.0 kV
Loading: 102.8% (Rating: 818 MVA)
DFAX: 7.7%
Pre-project loading: 101.7%
2. Facility: Ritter Park – Sugar Hill Substation 345.0 kV
Contingency: Hilltop – Piney Crk Project 500.0 kV
Loading: 100.8% (Rating: 1,622 MVA)
DFAX: 10.1%
Pre-project loading: 100.1%
3. Facility: Swan Vly – Trego 500.0 kV
Contingency: Wolfe – Barrett 500.0 kV line
Loading: 100.5% (Rating: 3,170 MVA)
DFAX: 7.1%
Pre-project loading: 100.2%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Weedpatch – Canal Road 230.0 kV	Reconductor line	\$7,386,000	1% (\$83,000)
Ritter Park – Sugar Hill 345.0 kV	Reconductor line	\$5,193,000	1% (\$59,000)
Swan Valley – Trego 500.0 kV	Reconductor line	\$39,639,000	1% (\$421,000)
	Total allocated to project		\$563,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.